

Hogfish

Species: *Lachnolaimus maximus*, Stock code: Hogfish - Florida Keys - East Florida

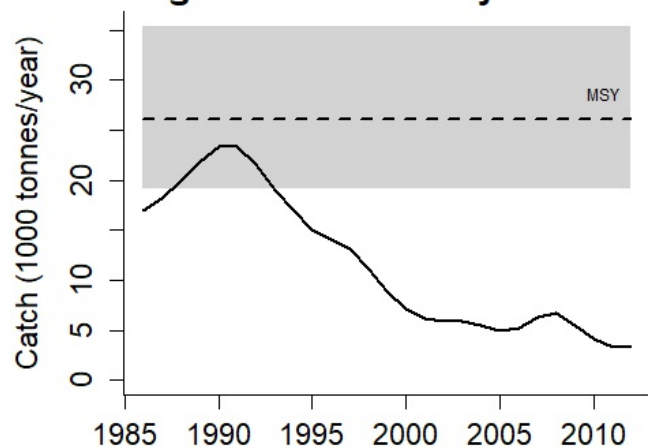
Region: Southeast Shelf

Marine Ecoregion: Florida Keys - East Florida

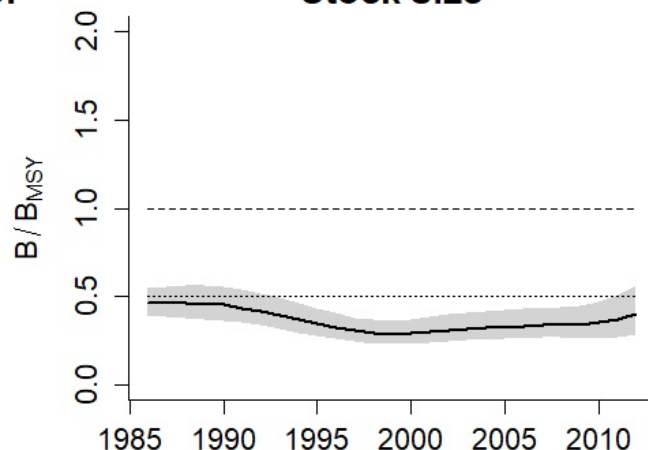
Reconstructed catch data used from years 1986 - 2012

For figure captions and method see <http://www.seaaroundus.org/cmsy-method>

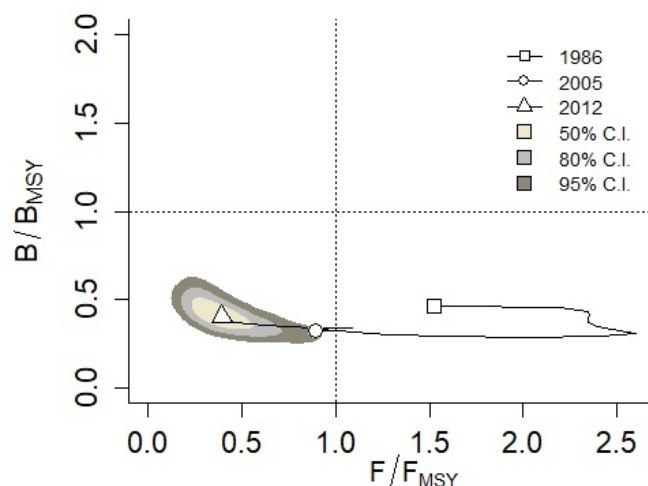
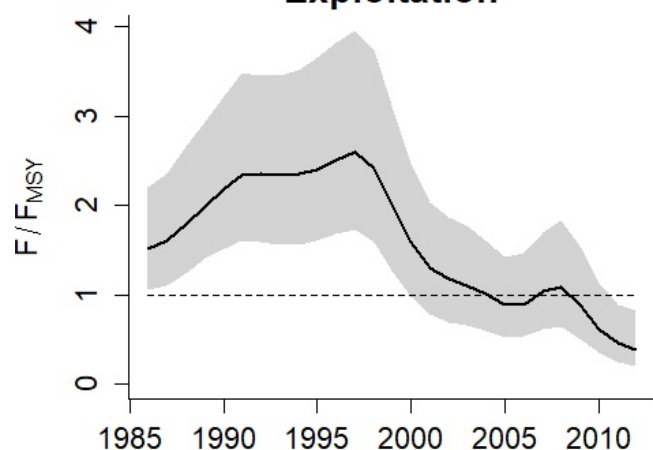
Catch Hogfish - Florida Keys - East Flori



Stock size



Exploitation



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Results for management (based on BSM analysis)

$F_{msy} = 0.0862$, 95% CL = 0.0535 - 0.134 (if $B > 1/2 B_{msy}$ then $F_{msy} = 0.5 r$)

$F_{msy} = 0.0697$, 95% CL = 0.0432 - 0.108 (r and F_{msy} are linearly reduced if $B < 1/2 B_{msy}$)

$MSY = 26$, 95% CL = 19.1 - 35.4; $B_{msy} = 304$, 95% CL = 190 - 525 (1000 tonnes)

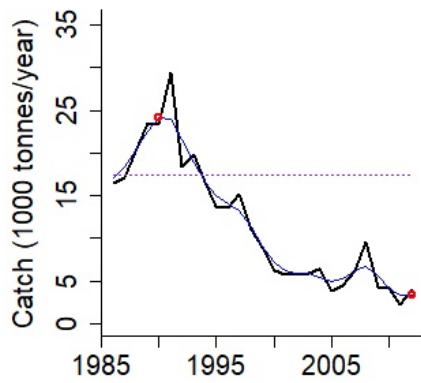
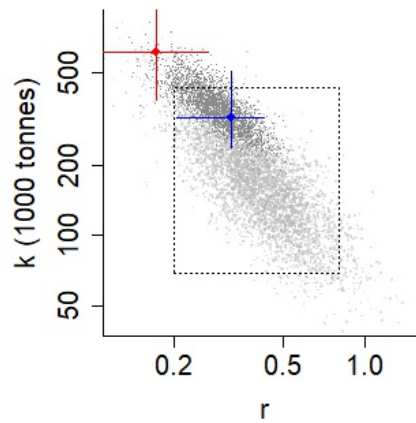
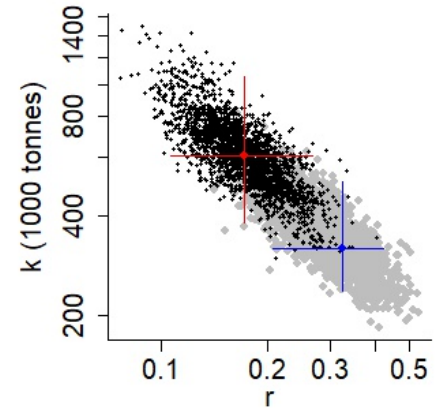
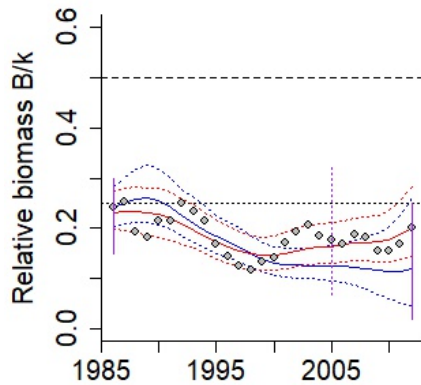
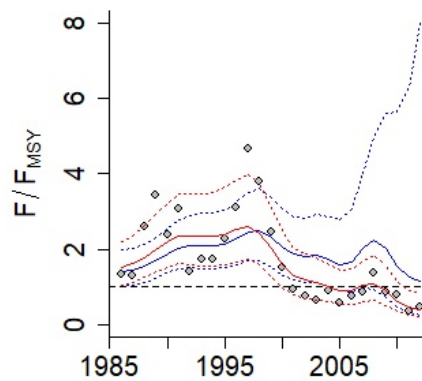
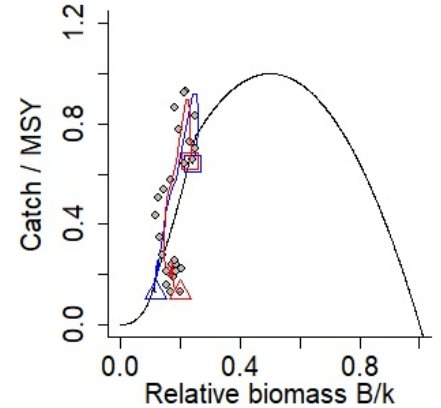
Biomass in last year = 123, 95% CL = 72.9 - 206 (1000 tonnes)

B/B_{msy} in last year = 0.404, 95% CL = 0.288 - 0.562

Fishing mortality in last year = 0.0274, 95% CL = 0.0153 - 0.0501

$F/F_{msy} = 0.392$, 95% CL = 0.207 - 0.835

Comment: NOAA Assessment model SS3. RF OK

A: Hogfish - Florida Keys - East Flor**B: Finding viable r-k****C: Analysis of viable r-k****D: Stock size****E: Exploitation rate****F: Equilibrium curve**

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Results of CMSY analysis conducted in JAGS

$r = 0.323$, 95% CL = 0.206 - 0.424; $k = 319$, 95% CL = 237 - 507 (1000 tonnes)

MSY = 25.7, 95% CL = 20.1 - 33.8 (1000 tonnes/year)

Relative biomass last year = 0.119 k , 95% CL = 0.0456 - 0.259

Exploitation $F/(r/2)$ in last year = 1.26

Results from Bayesian Schaefer model using catch and CPUE

$r = 0.172$, 95% CL = 0.107 - 0.267; $k = 608$, 95% CL = 381 - 1050

r - k log correlation = -0.795

MSY = 26, 95% CL = 19.1 - 35.4 (1000 tonnes/year)

Relative biomass in last year = 0.119 k , 95% CL = 0.0456 - 0.259

Exploitation $F/(r/2)$ in last year = 0.637

$q = 3.28$, 95% CL = 1.98 - 5.19

Prior range of q = 1.16 - 28.6

Relative abundance data type = CPUE

Prior initial relative biomass = 0.15 - 0.3 expert

Prior intermediate relative biomass = 0.0678 - 0.323 in year 2005 default

Prior final relative biomass = 0.02 - 0.25 expert

Prior range for r = 0.2 - 0.8 default, prior range for k = 69.1 - 428 (1000 tonnes) default

Source for relative biomass:

NOAA Fisheries. (2020). Assessment Summary Data. Retrieved from www.st.nmfs.noaa.gov/stocksmart. 12/13/2020